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"ADVANCEMENTS IN NEXT-GENERATION INTELLIGENT SOLUTION"

Paper Title:

MonBarta: Deep Dive into Psychological Diagnosis of Mental Health Conversations Using LLM

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Authors:

Saidur Rahman Sujon
Ahmadul Karim Chowdhury
Ashek Seum
Anup Kumar Sutradhar
Faisal Muhammad Shah

Affiliation:

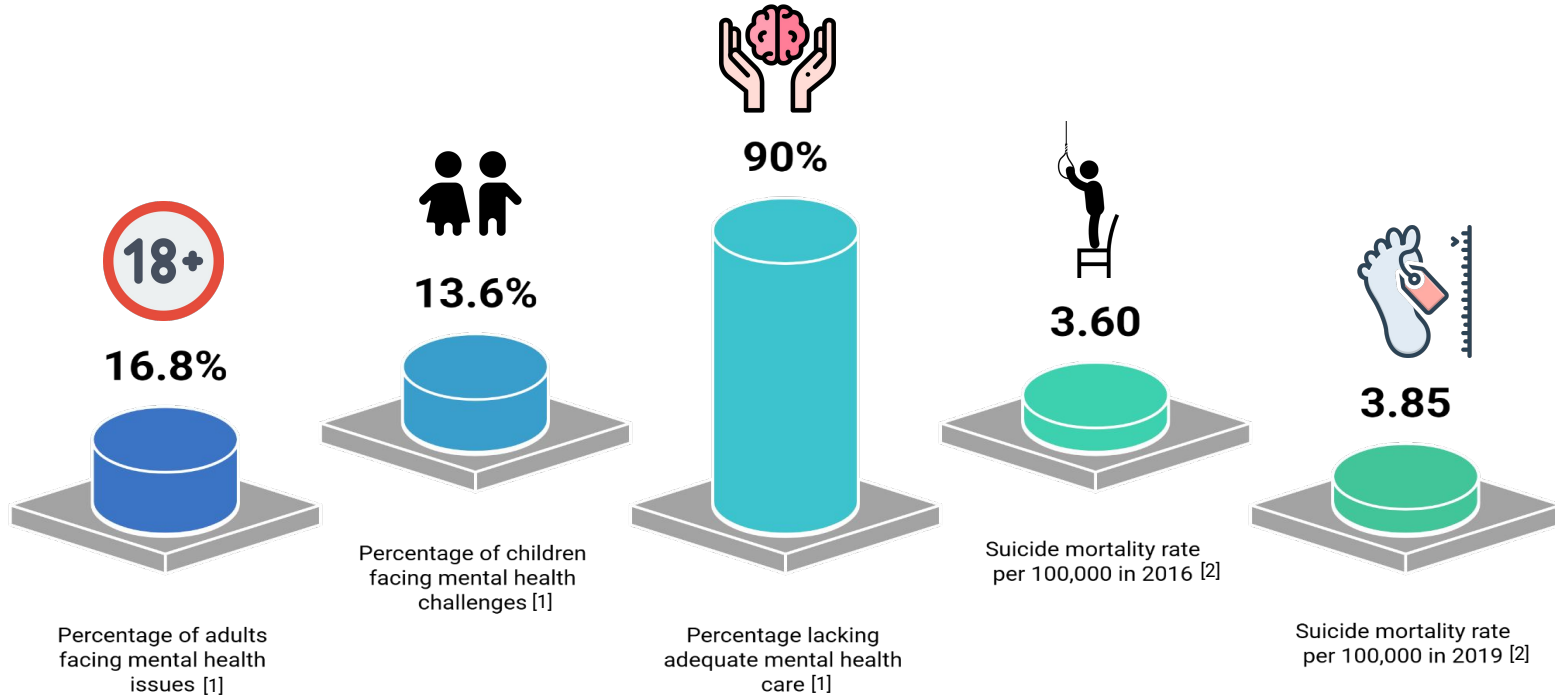
Dept. of Computer Science and Engineering,
Ahsanullah University of Science and Technology, Bangladesh

01	INTRODUCTION	Mental Health Statistics in Bangladesh, Research Questions, Research Objectives	
02	RELATED WORK	Recent advances in mental health analysis	
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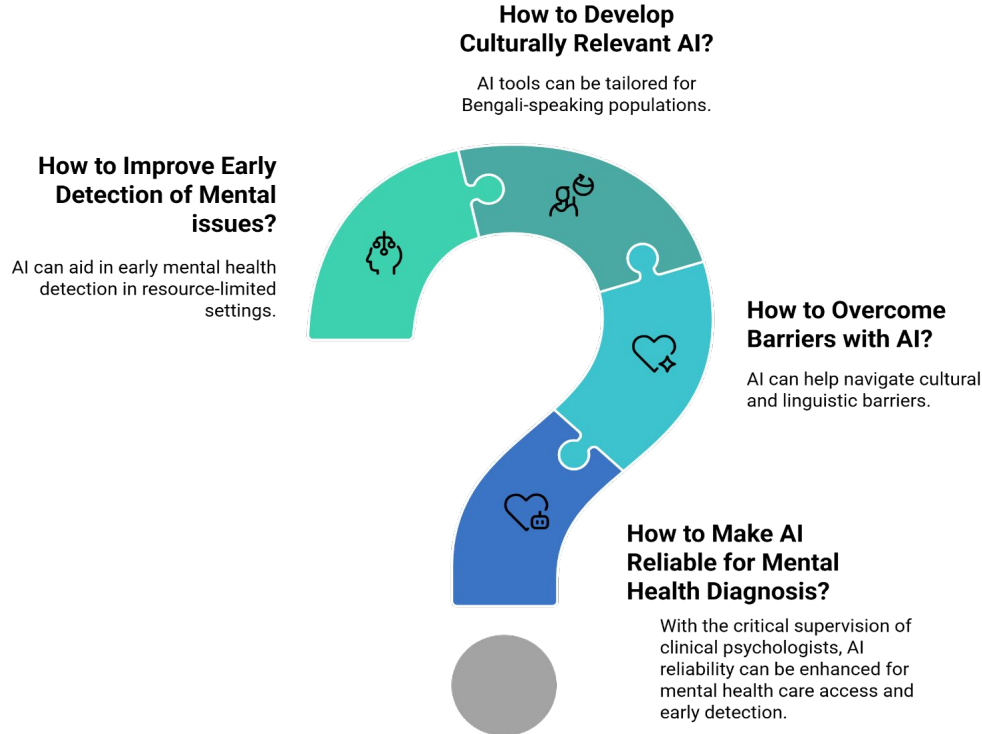
Introduction

Mental Health Statistics in Bangladesh



Research Questions

How can AI address Mental Health Challenges in Bangladesh?



Research Objective



Develop Resources

Develop Bengali mental health resources with a clinical psychologist for early intervention



Create Dataset

Create a dataset with synthetic Bengali conversations, validated by psychologists. Key attributes: Conversation, Symptoms, Confidence score



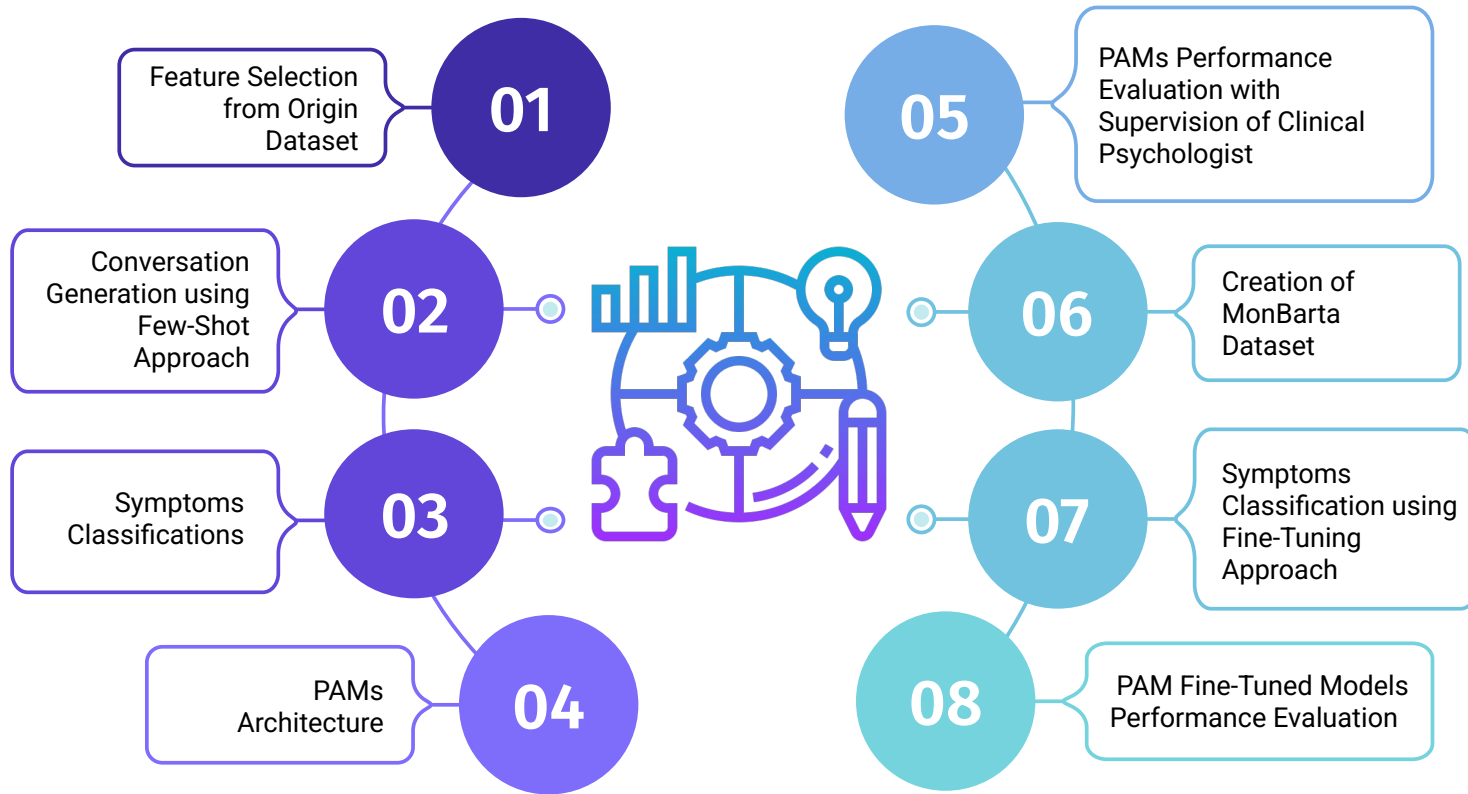
Train Models

Fine-tune models to achieving higher accuracy in condition classification

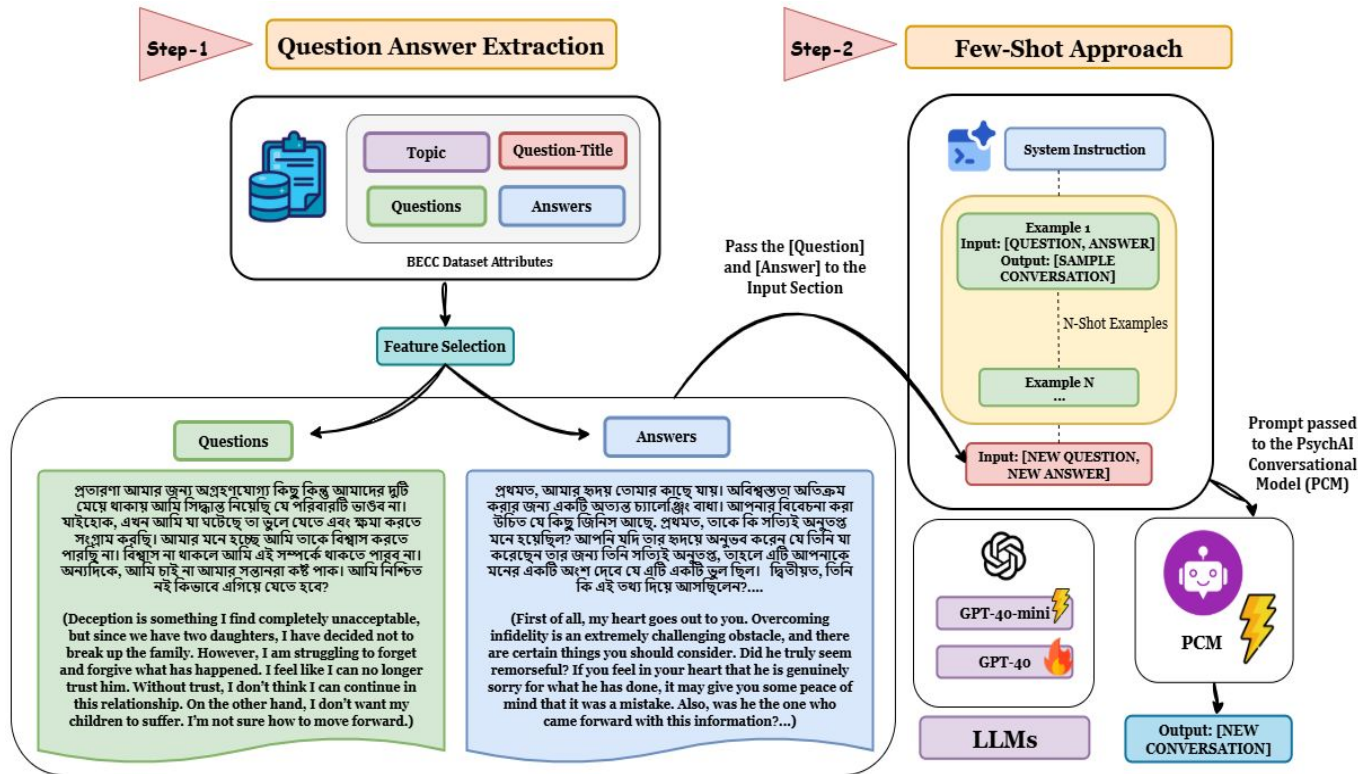
Related Works

Title	Methodology	Result
ChatCounselor A Large Language Models for Mental Health Support [3]	- Trained on Psych8k dataset from 260 in-depth interviews. - Constructed dataset involved licensed psychological counselors	- Enhanced performance with fine-tuning on a small amount of high-quality, domain-specific data.
A Shoulder to Cry on: Towards A Motivational Virtual Assistant for Assuaging Mental Agony [4]	- Utilized the MotiVAtte dataset (7k dyadic conversations from peer support platform). - Employed attention-based BERT classifier and sentiment-driven Reinforcement Learning (RL) for dialogue generation.	- BERT embeddings yielded better results than GloVe - Sentiment-based RL improved generation of positive responses.
Pioneering Mental Health Conversations Utilizing Large Language Models for Talk Therapy [5]	- Fine-tuned Gemma 2B v1.1 Instruct model - Employed LoRA to optimize model performance	- Fine-tuning improved the model's ability to generate empathetic and personalized mental health responses
BengaliEmpatheticConversationsCorpus: A Comprehensive Bengali Language Dataset for Mental Health Counseling [6]	- Created a corpus of 38,235 query-answer pairs, focusing on empathetic conversations	- First and largest corpus for empathetic counseling in Bengali.
MentalArena Self-play Training of Language Models for Diagnosis and Treatment of Mental Health Disorders [7]	- Utilized a self-play framework where a model acts as both patient and therapist - Generated 18k samples of data, validated using GPT-4o	- Models fine-tuned with MentalArena outperformed GPT-4o, GPT-3.5-turbo, and Llama-3-8b
Mental-LLM Leveraging Large Language Models for Mental Health Prediction via Online Text Data [8]	- Utilized four high-quality human-annotated social media datasets - Datasets included stress, depression, suicide ideation, and suicide risk prediction tasks.	- Instruction finetuning on Alpaca and FLAN-T5 models significantly improved performance, surpassing GPT-3.5/4.

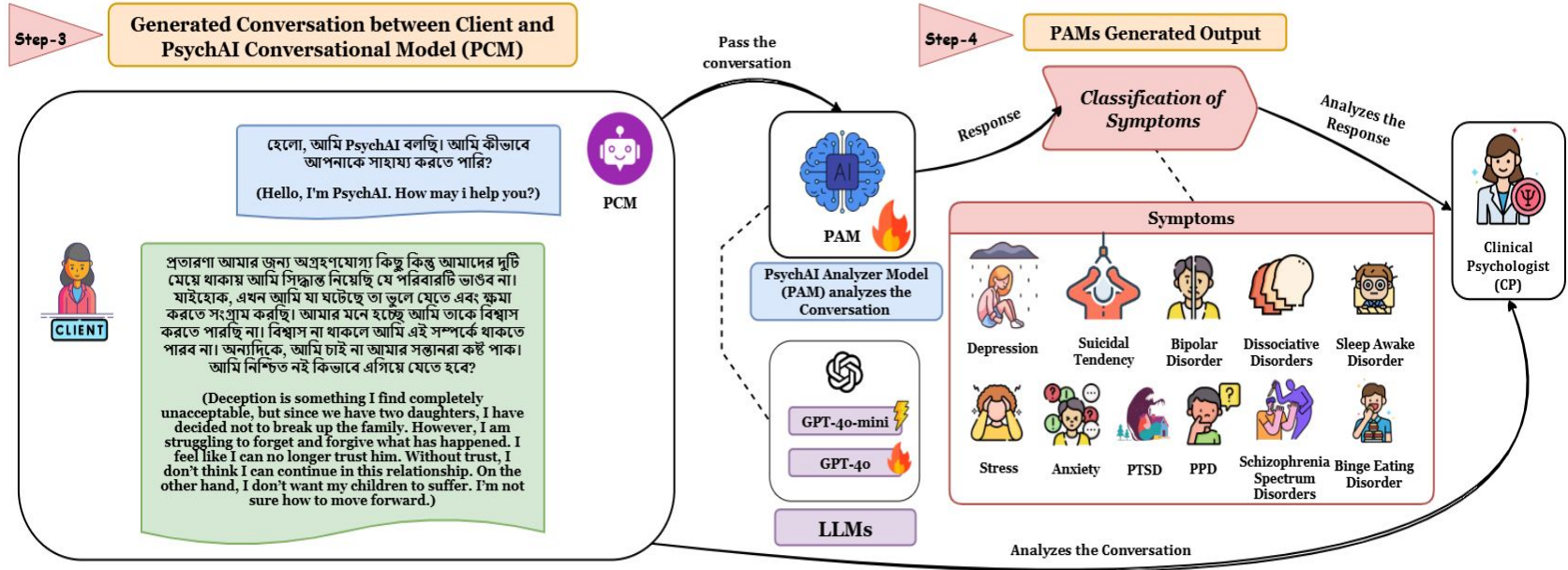
Methodology



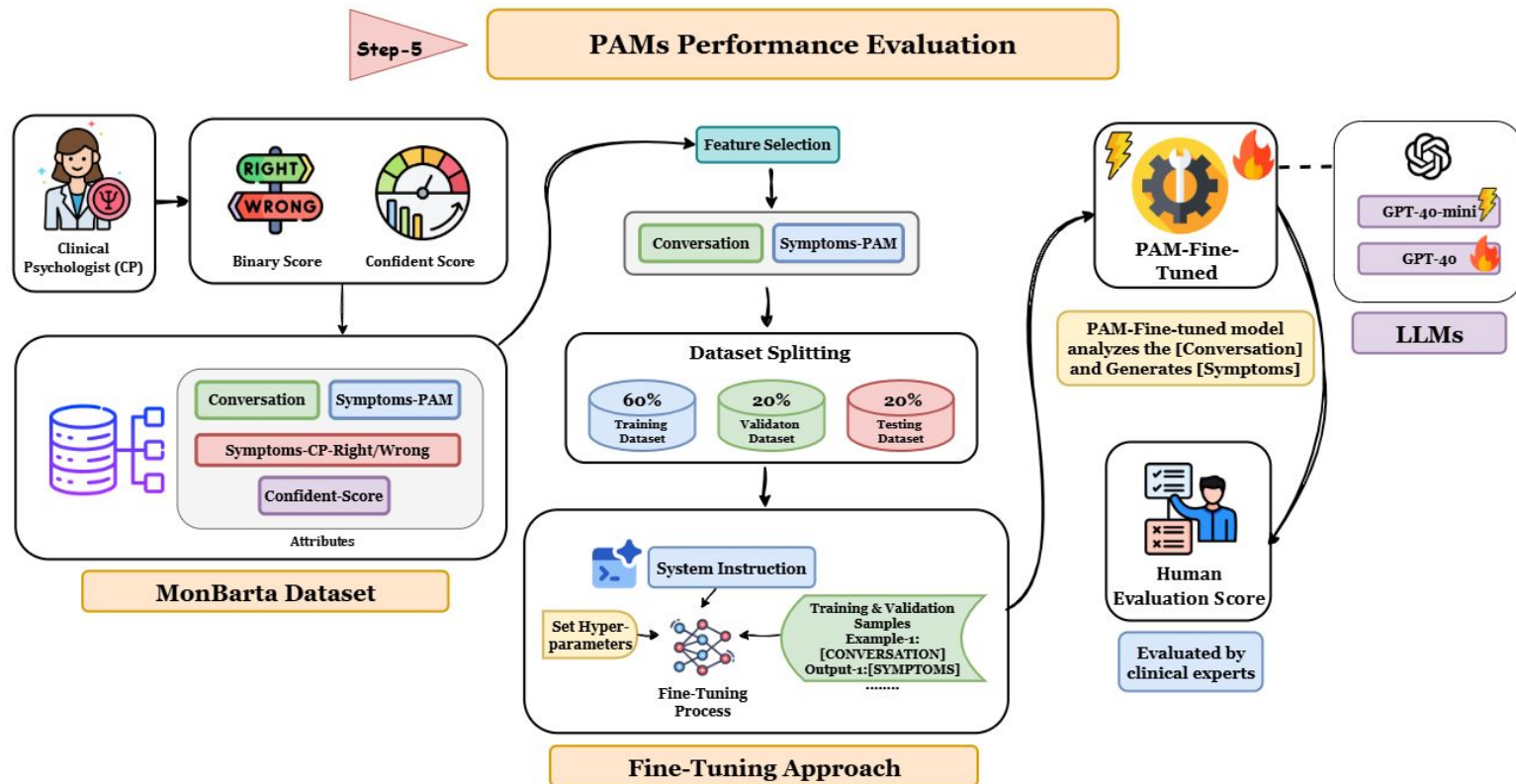
PsychAI Conversational Model (PCM) Architecture



PsychAI Analyzer Model (PAM) Architecture



PAM Fine-Tuned Model Architecture & Evaluation Criteria



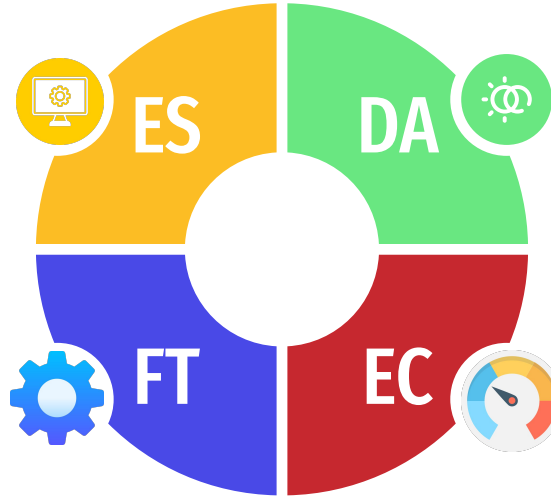
Experimental Analysis

Environment Settings

Google Colab Pro (T4 GPU)
RAM 32GB
OpenAI API

PAM Fine-Tuned Models

Hyperparameters
(batch size = 1, learning rate multiplier=1.8, epoch = 5)



MonBarta Dataset Analysis

200 samples
Attributes: Conversations,
Symptoms-PAM, Symptoms-CP,
and Confidence-Score

Evaluation Criteria

Average Human Evaluation
Score (AHES) by clinical
experts (*5) → %

MonBarta Dataset Analysis

TABLE I. DISTRIBUTION OF TOKENS ACROSS DATASET SPLITS

Type	Tokens	Overall (%)
Train	93,452	60
Validation	28,252	20
Test	33,530	20
Total	155,234	100

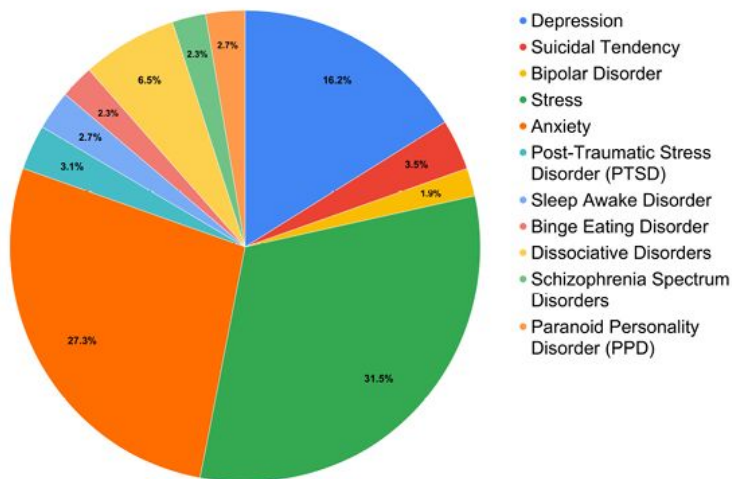


Fig. 2. The percentage distribution of all symptoms in the MonBarta dataset.

TABLE II. SUB-LEVELS OF EVALUATION CRITERIA FOR MONBARTA DATASET

Levels	HES
1	Unacceptable
2	Needs Improvement
3	Satisfactory
4	Good
5	Outstanding



Fig. 3. A word-cloud representation of the most used words in conversations in MonBarta Dataset.

PAM Fine-Tuned Models Training-Validation Graph

Platform: OpenAI
Steps: 0 to 300 (PAM-P1) & 0 to 570 (PAM-P2)
Total Trained Tokens: 1,656,145 across all epochs

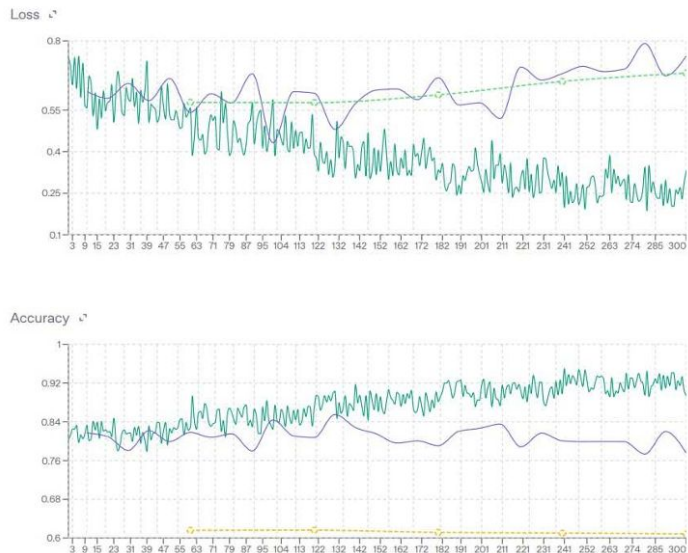


Fig. 4. Training and validation performance of PAM-P1 (fine-tuned GPT-4o-mini model) showing loss reduction and accuracy trends across epochs.

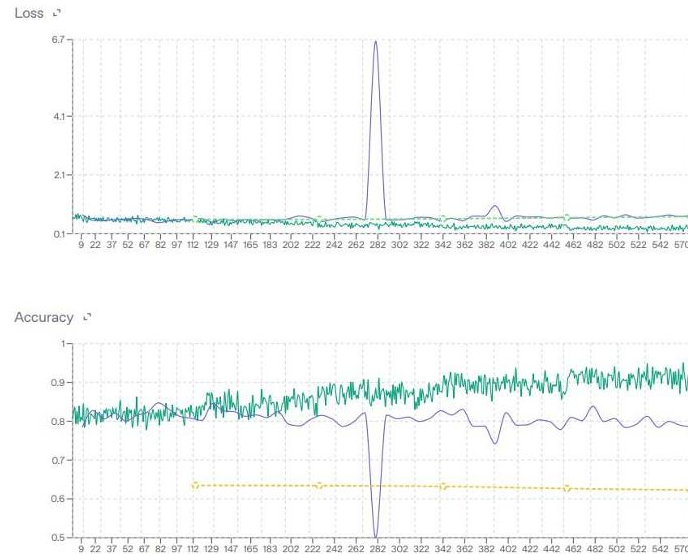


Fig. 5. Training and validation performance of PAM-P2 (fine-tuned GPT-4o model) showing loss reduction and accuracy trends across epochs.

Results Analysis and Discussion



Performance Metrics of PAM

TABLE III. PERFORMANCE METRICS OF PAM MODEL VARIANTS. AHES: AVERAGE HUMAN EVALUATION SCORE

Model	Base Model	Approach	AHES (* /5)	AHES (%)
PAM-P1	GPT-4o-mini	Fine-Tuned	4.3421	86.842
PAM-P2	GPT-4o	Fine-Tuned	4.2802	85.604



Comparing with other existing datasets

TABLE IV. SUMMARY OF MENTAL HEALTH DATASETS WITH ACCURACY, AHES (AVERAGE HUMAN EVALUATION SCORE), AND CLASSIFICATION TYPES USING THE PAM-P1 MODEL

Dataset	Accuracy (%)	AHES (* /5)	Classification Type
DR [24]	85.00	3.81	Binary (Depression, Not Depression)
dreaddit [25]	85.71	3.86	Binary (Stress, Not Stress)
SWMH [26]	84.61	3.75	Multiclass (Anxiety, Depression, Bipolar Disorder)
Counsel Chat [14]	88.89	4.17	Multiclass (Stress, Anxiety, Depression, Binge Eating Disorder, PTSD)

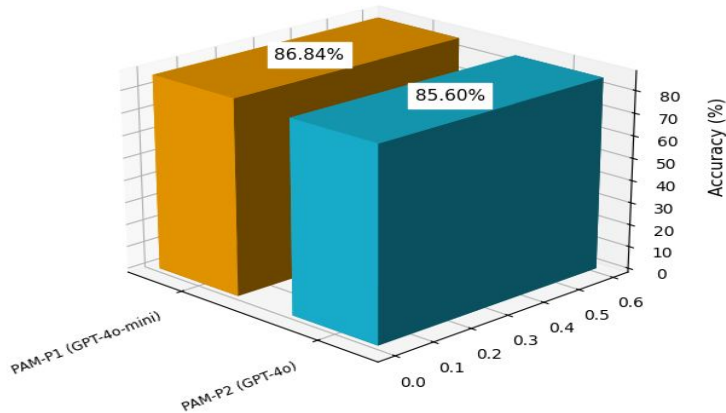


Results Analysis and Discussion



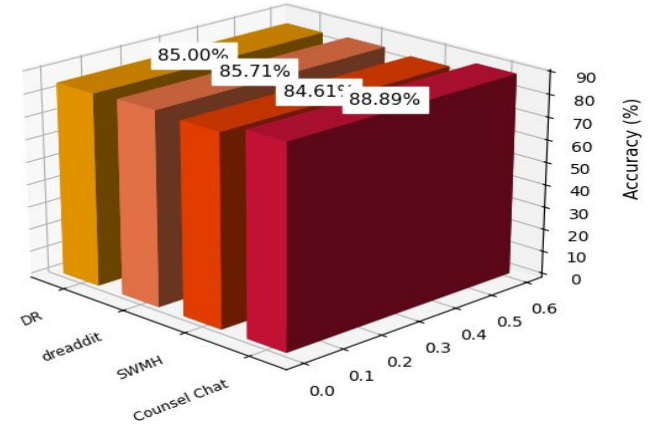
Performance Metrics of PAM

Accuracy (%) for PAM Model Variants



Comparing with other existing datasets

Accuracy (%) for Mental Health Datasets (using PAM-P1)



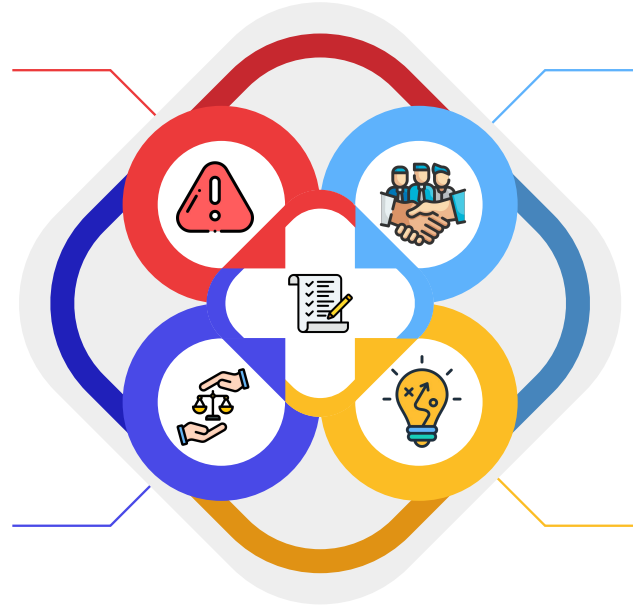
Conclusion

Limitation

Biases related to depression, stress, and anxiety
Hallucinated symptoms (e.g., social stress, low self-esteem, trust issues)

Ethical Consideration

Validated by expert psychologists,
Anonymized sensitive details,
Strictly prohibited any medical suggestions, Data privacy



Acknowledgement

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Future Work

Hallucination and Bias mitigation by exploring advanced LLMs prompt engineering approaches

References

1. <https://thefinancialexpress.com.bd/health/only-260-psychiatrists-for-bangladeshs-mental-health-patients-1602323649>
2. <https://www.who.int/publications/m/item/mental-health-atlas-bgd-2020-country-profile>
3. <https://arxiv.org/pdf/2309.15461>
4. https://aclanthology.org/2022.naacl-main.174.pdf?fbclid=IwAR2sq6Ubv5F9bVzXiHj2d3HaMwYq72dhcwP1M_5sD3oyvYez7Pp7HejWxNg
5. <https://www.techrxiv.org/users/834271/articles/1228639-pioneering-mental-health-conversations-utilizing-large-language-models-for-talk-therapy>
6. <https://data.mendeley.com/datasets/b3j5tkswm7/1>
7. https://www.researchgate.net/publication/384770780_MentalArena_Self-play_Training_of_Language_Models_for_Diagnosis_and_Treatment_of_Mental_Health_Disorders
8. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11806945/>



Thank You for Your Attention !
Any Questions 